Glossary Appendix: Low-Rank Structure Is Geometry

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This notebook is an **optional companion** to the main tutorial.

Use it when you encounter a term that is unfamiliar. Mathematical and AI/interpretability vocabulary are organized into topic clusters, followed by a cross-language map showing where the two vocabularies meet.

> **Two recurring cautions**
>
> **coordinates ≠ intrinsic object learned factor / dictionary atom ≠ validated semantic concept**

Mathematics

46 terms

I. Tensor basics

Tensor

TENSOR BASICS

A multidimensional array $\mathcal{X} \in \mathbb{R}^{n_1 \times \cdots \times n_d}$. A matrix is the order-2 case. The notebooks preserve meaningful modes instead of always flattening them.

▶ **Sources [KB09]**

Mode

TENSOR BASICS

One axis of a tensor. A mode may represent samples, spatial positions, features, time, or another scientifically meaningful variable class.

▶ **Sources [KB09]**

Matricization / unfolding

TENSOR BASICS

A rearrangement of a tensor into a matrix by choosing one or more modes as matrix axes. Flattening can enable matrix methods while merging distinct mode semantics.

▶ **Sources [KB09]**

Outer product

TENSOR BASICS

For vectors a, b, c, the tensor $a \otimes b \otimes c$ has entries $(a \otimes b \otimes c)_{ijk} = a_i b_j c_k$. Every CP rank-one term is an outer product of one vector from each mode.

▶ **Sources [KB09]**

Rank-one tensor

TENSOR BASICS

A tensor that can be written as one outer product $a^{(1)} \otimes \cdots \otimes a^{(d)}$. It is the basic component of CP decomposition and the Segre model.

▶ **Sources [KB09] [BV18]**

Matrix rank

TENSOR BASICS

The dimension of the column space (equivalently, row space) of a matrix. Matrix rank should not be confused with CP rank or multilinear rank.

II. Decomposition models

CP decomposition

DECOMPOSITION MODELS

A model $\mathcal{X} \approx \sum_{r=1}^{R} \lambda_r a_r^{(1)} \otimes \cdots \otimes a_r^{(d)}$. CP is the main example where factor vectors are coordinates while the reconstructed tensor is the represented object.

▶ **Sources [KB09] [BV18]**

CP rank

DECOMPOSITION MODELS

The smallest number of rank-one tensors whose sum equals a tensor exactly. It differs from matrix rank and Tucker multilinear rank.

▶ **Sources [KB09]**

Factor matrix

DECOMPOSITION MODELS

In CP, vectors from one mode are collected as columns of a factor matrix. Its columns are coordinates and can change under CP scaling or permutation equivalences.

▶ **Sources [KB09]**

Tucker decomposition

DECOMPOSITION MODELS

A model $\mathcal{X} \approx \mathcal{G} \times_1 U_1 \times_2 \cdots \times_d U_d$ with a core tensor and mode factors. Tucker emphasizes mode-wise subspaces and their interactions rather than a sum of rank-one terms.

▶ **Sources [KB09] [DLV00]**

Core tensor

DECOMPOSITION MODELS

The smaller tensor $\mathcal{G}$ in a Tucker representation that records interactions among selected mode coordinates. Individual core entries are basis-dependent; the mode subspaces can be more invariant than a particular basis.

▶ **Sources [KB09]**

Multilinear rank

DECOMPOSITION MODELS

The tuple $(r_1, \ldots, r_d)$ of ranks of the tensor's mode unfoldings. It is the rank notion naturally associated with Tucker models.

▶ **Sources [DLV00]**

HOSVD

DECOMPOSITION MODELS

Higher-Order SVD constructs Tucker mode subspaces from singular vectors of tensor unfoldings. It gives an operational way to inspect multilinear rank and compress a tensor.

▶ **Sources [DLV00]**

Block-Term Decomposition / BTD

DECOMPOSITION MODELS

A sum of structured multilinear-rank blocks rather than only rank-one terms. A BTD component can represent within-component interactions that are not separable across modes.

▶ **Sources [DL08]**

SVD

DECOMPOSITION MODELS

For a matrix X, $X = U\Sigma V^\top$. It provides an optimal low-rank approximation for a chosen matricization but can merge mode semantics after flattening.

▶ **Sources [KB09]**

III. Geometry and equivalence

Subspace

GEOMETRY AND EQUIVALENCE

A subset of a vector space closed under addition and scalar multiplication. Low-rank matrix and Tucker methods often identify a subspace more naturally than an individually meaningful basis vector.

▶ **Sources [AMS08]**

Orthonormal basis

GEOMETRY AND EQUIVALENCE

A basis whose vectors have unit norm and are mutually orthogonal. An orthonormal matrix is a convenient coordinate system for a subspace, but different bases can span the same subspace.

Manifold

GEOMETRY AND EQUIVALENCE

A space that locally resembles Euclidean space and supports tangent-space calculus. Several low-rank model spaces are naturally treated as manifolds for optimization.

▶ **Sources [AMS08]**

Stiefel manifold

GEOMETRY AND EQUIVALENCE

$\mathrm{St}(n, r) = \{U \in \mathbb{R}^{n\times r} : U^\top U = I_r\}$, the set of matrices with orthonormal columns. It appears when learning orthonormal bases or subspace coordinates, as in StelLA.

▶ **Sources [AMS08] [LIS25]**

Segre manifold / variety

GEOMETRY AND EQUIVALENCE

The geometric model of rank-one tensors obtained from outer products. A CP tensor is a sum of Segre components.

▶ **Sources [BV18]**

Fixed-rank manifold

GEOMETRY AND EQUIVALENCE

The smooth manifold of matrices having a fixed rank, away from rank-changing boundaries. It allows direct optimization of a low-rank matrix object without introducing a redundant factorization.

▶ **Sources [AMS08] [BIA25]**

Coordinates vs represented object

GEOMETRY AND EQUIVALENCE

A parameterization gives coordinates to an object; different coordinates may represent exactly the same object. This is the central distinction of the tutorial: factors are coordinates, while the reconstructed matrix or tensor is the object.

▶ **Sources [BV18]**

Gauge freedom / equivalence

GEOMETRY AND EQUIVALENCE

A family of coordinate transformations that leaves the represented object unchanged.

▶ **Why it matters**

Scaling ambiguity

GEOMETRY AND EQUIVALENCE

A reciprocal rescaling of factors can preserve the same rank-one tensor, e.g. $(\alpha a) \otimes (\beta b) \otimes ((\alpha\beta)^{-1} c)$. Factor norms can change substantially while reconstruction stays unchanged.

▶ **Sources [KB09]**

Permutation ambiguity

GEOMETRY AND EQUIVALENCE

Reordering CP components does not change their sum. Two CP outputs should be aligned before column-by-column comparison.

▶ **Sources [KB09]**

Identifiability

GEOMETRY AND EQUIVALENCE

A model is identifiable when the represented object determines its parameters uniquely after accepted equivalences are accounted for.

▶ **Interpretation boundary**

▶ **Sources [KB09] [DL08]**

IV. Optimization

Nonconvex optimization

OPTIMIZATION

Optimization in which the objective or feasible set does not have global convexity guarantees. Low-rank factorizations can have initialization-dependent trajectories and multiple stationary regions.

Initialization

OPTIMIZATION

The starting coordinates or starting low-rank object supplied to an iterative solver. Different starts can produce different optimization trajectories even for the same target and model.

▶ **Sources [KB09]**

ALS

OPTIMIZATION

Alternating Least Squares updates one factor block at a time while keeping the others fixed. It is simple and effective, but its local least-squares systems can become poorly conditioned.

▶ **Sources [KB09]**

RALS

OPTIMIZATION

Regularized ALS adds a stabilizing term to alternating updates. It illustrates that solver design can change behavior near difficult regions without changing the model family.

Riemannian optimization

OPTIMIZATION

Optimization performed on a manifold using tangent-space gradients and manifold-compatible updates. It respects the geometry of low-rank model spaces instead of treating all coordinates as unconstrained Euclidean parameters.

▶ **Sources [AMS08]**

Tangent space

OPTIMIZATION

The linear space of infinitesimal feasible directions at a point on a manifold. Riemannian gradients and search directions live in tangent spaces.

▶ **Sources [AMS08]**

Retraction

OPTIMIZATION

A map that sends a tangent-space step back to the manifold and agrees locally with the manifold geometry. It turns a tangent search direction into a new feasible point.

▶ **Sources [AMS08]**

RGD

OPTIMIZATION

Riemannian Gradient Descent follows the negative Riemannian gradient and retracts each step. It is a basic geometry-aware alternative to Euclidean or alternating updates.

▶ **Sources [AMS08]**

RCG

OPTIMIZATION

Riemannian Conjugate Gradient combines the current gradient with transported information from previous directions. It provides a geometry-aware acceleration strategy beyond steepest descent.

▶ **Sources [AMS08]**

Levenberg–Marquardt / LM

OPTIMIZATION

A damped Gauss–Newton-type method for nonlinear least-squares problems. It provides a curvature-aware strategy for low-rank reconstruction.

V. Failure and stability

Conditioning

FAILURE AND STABILITY

Sensitivity of a solution or representation to small perturbations of the input or represented object.

▶ **Interpretation boundary**

▶ **Sources [BV18]**

Condition number

FAILURE AND STABILITY

A numerical quantity that measures sensitivity; for nonsingular M, one common form is $\kappa(M) = \|M\|\|M^{-1}\|$. Different condition numbers diagnose different objects; ALS subproblem conditioning and geometric CP conditioning should not be conflated.

▶ **Sources [BV18]**

Relative reconstruction error

FAILURE AND STABILITY

$\|X - \widehat{X}\|_F / \|X\|_F$ (and analogously for tensors). It measures object-level fit, not uniqueness, stability, or semantic validity.

▶ **Sources [KB09]**

Overparameterization / excessive rank

FAILURE AND STABILITY

Using more latent components than are needed to represent the dominant structure. Extra components can split or duplicate structure and complicate interpretation.

Rank choice

FAILURE AND STABILITY

The selected number of components or latent dimensions used by a model.

▶ **Interpretation boundary**

Component collision

FAILURE AND STABILITY

Two normalized rank-one components become nearly indistinguishable as represented rank-one tensors, up to accepted scale/sign conventions.

▶ **Why it matters**

▶ **Sources [BV18]**

Degeneracy

FAILURE AND STABILITY

A tensor-approximation pathology in which component norms can grow while their sum remains bounded, often through near cancellation. A small represented tensor can hide large, fragile component coordinates.

▶ **Sources [KB09] [BV18]**

Cancellation

FAILURE AND STABILITY

Large terms with opposing contributions combine to produce a much smaller sum. It is a separate pathology from component collision, even though the two may occur together.

Swamp / swamp-like stagnation

FAILURE AND STABILITY

An extended region of very slow objective improvement during an iterative tensor-decomposition method. A plateau is an observation, not an explanation: collision, degeneracy, initialization, rank choice, and model mismatch are possible contributors.

▶ Sources [KB09]

Model mismatch

FAILURE AND STABILITY

The chosen model family does not adequately represent the structure that generated the data. Poor fit is not always an optimization failure; the structural assumption itself may be wrong.

AI & Interpretability

71 terms

I. Neural representations

Neural representation

NEURAL REPRESENTATIONS

The internal activation state produced by a neural network for an input at a chosen layer or module. Low-rank methods are applied to collections of such states to test hypotheses about recurring structure.

▶ Sources [KIM18] [GHO19] [FEL23]

Activation

NEURAL REPRESENTATIONS

The numerical output of a neuron, channel, token position, or other internal unit after a layer processes an input. In the CRAFT-style lab, each image crop produces an activation vector.

▶ Sources [FEL23]

Activation space

NEURAL REPRESENTATIONS

The vector space in which a chosen layer's activation vectors live. CAVs, NMF directions, and SAE decoder directions can all be discussed as directions in an activation space.

▶ **Sources [KIM18] [CUN23]**

Latent representation

NEURAL REPRESENTATIONS

A lower-dimensional or internal representation that is not identical to the original input. CP, Tucker, NMF, SVD, and autoencoders all introduce latent coordinates under different assumptions.

Representation learning

NEURAL REPRESENTATIONS

Learning useful internal variables or coordinates from data rather than specifying all features by hand. The tutorial compares several structured representation families and asks what their coordinates can legitimately mean.

Feature

NEURAL REPRESENTATIONS

An overloaded term that may mean an input variable, neuron, activation direction, or learned latent factor. The tutorial avoids assuming that every learned factor is automatically a semantic feature.

Feature direction

NEURAL REPRESENTATIONS

A direction in activation space associated with a recurring learned pattern. Dictionary atoms, CAVs, and SAE decoder columns can all be treated as candidate feature directions, with different training assumptions.

▶ **Sources [KIM18] [CUN23]**

II. Dictionary representations

Dictionary learning

DICTIONARY REPRESENTATIONS

Learning a set of reusable basis elements D such that observations can be approximated as combinations $x_i \approx D\alpha_i$. It is the broader representation-learning idea behind learned feature dictionaries; NMF and sparse autoencoders impose different additional structures.

▶ **Sources [MBPS09]**

Dictionary atom

DICTIONARY REPRESENTATIONS

One learned basis element or direction in a dictionary.

▶ **Interpretation boundary**

▶ **Sources [MBPS09]**

Sparse coding

DICTIONARY REPRESENTATIONS

Representing each observation with only a small number of active dictionary coefficients. Sparsity and nonnegativity are distinct assumptions: a code can be sparse without being nonnegative and vice versa.

▶ **Sources [MBPS09]**

Sparsity

DICTIONARY REPRESENTATIONS

The property that most entries of a representation are zero or inactive. Sparse representations encourage each observation to use only a small subset of learned features.

▶ **Sources [MBPS09] [CUN23]**

Sparsity penalty

DICTIONARY REPRESENTATIONS

A regularization term or constraint that encourages few active latent variables, such as an ℓ_1 penalty or related activation penalty. In SAEs it trades reconstruction fidelity against sparse feature usage.

Autoencoder

DICTIONARY REPRESENTATIONS

A model trained to reconstruct an input through an intermediate latent code: $z = f_{enc}(x)$ and $\hat{x} = f_{dec}(z)$. It provides the basic architecture from which sparse autoencoders are built.

Encoder

DICTIONARY REPRESENTATIONS

The mapping f_{enc} that converts an input or activation vector into latent coordinates. In an SAE, the encoder determines which learned latent features are active.

▶ **Sources [CUN23]**

Decoder

DICTIONARY REPRESENTATIONS

The mapping f_{dec} that reconstructs the input from the latent code. For a linear SAE decoder, decoder columns are commonly treated as learned feature directions or dictionary atoms.

▶ **Sources [CUN23]**

Latent code

DICTIONARY REPRESENTATIONS

The intermediate representation z produced by an encoder. In an SAE, sparsity is imposed on this code so that only a small number of features activate for each input.

Sparse autoencoder / SAE

DICTIONARY REPRESENTATIONS

An autoencoder trained to reconstruct activations through a latent code that is encouraged to be sparse.

▶ **Interpretation boundary**

▶ **Sources [CUN23] [BRI23] [TEM26]**

Overcomplete dictionary

DICTIONARY REPRESENTATIONS

A dictionary with more learned atoms/features than the dimensionality of the input representation. Overcomplete feature sets are used to model the hypothesis that networks may represent more features than neurons/dimensions via superposition.

▶ **Sources [CUN23] [BRI23]**

Feature activation

DICTIONARY REPRESENTATIONS

The coefficient or latent value indicating how strongly a learned feature is active for one example. High activation means presence under that representation; it does not by itself establish importance or causal relevance.

Dead feature / dead latent

DICTIONARY REPRESENTATIONS

A learned latent feature that rarely or never activates on the data distribution used to evaluate it. Dead features indicate unused dictionary capacity and are a practical SAE training diagnostic.

Monosemanticity

DICTIONARY REPRESENTATIONS

The ideal that one learned feature corresponds to one coherent semantic pattern rather than several unrelated meanings. SAE work investigates whether sparse dictionaries yield more monosemantic features than individual neurons.

▶ **Sources [CUN23] [BRI23] [TEM26]**

Polysemanticity

DICTIONARY REPRESENTATIONS

A neuron or learned feature responds to several semantically distinct patterns. It motivates attempts to recover cleaner latent directions than individual neurons provide.

▶ **Sources [CUN23] [BRI23]**

Superposition

DICTIONARY REPRESENTATIONS

The hypothesis that a network represents more features than available neurons/dimensions by encoding features in non-orthogonal directions. Sparse autoencoders and dictionary learning are used to search for those latent directions.

▶ **Sources [CUN23] [BRI23]**

Feature splitting

DICTIONARY REPRESENTATIONS

A phenomenon where a semantic pattern that might be described as one feature at one scale is represented by several more specific learned features at another scale or model capacity. It warns that increasing dictionary size can change the vocabulary rather than simply refine reconstruction.

▶ **Sources [TEM26]**

NMF

DICTIONARY REPRESENTATIONS

A constrained factorization $A \approx UW^\top$ with $U, W \geq 0$.

▶ **Interpretation boundary**

▶ **Sources [LS99] [FEL23]**

III. Concept-based interpretability

Concept

CONCEPT-BASED INTERPRETABILITY

A higher-level pattern used to describe model representations or behavior in human-relevant terms. A learned direction is first a candidate concept; semantic meaning requires additional evidence.

▶ **Sources [KIM18] [GHO19]**

Concept-based interpretability

CONCEPT-BASED INTERPRETABILITY

Explaining neural networks using higher-level concepts rather than only individual input features or neurons. CRAFT is the main case study in Lab 4.

▶ **Sources [KIM18] [GHO19] [FEL23]**

Concept discovery

CONCEPT-BASED INTERPRETABILITY

Searching for recurring structure in neural activations that may correspond to meaningful higher-level patterns, often without pre-specifying all concepts. Discovery yields candidate explanations, not ground-truth semantics.

▶ **Sources [GHO19] [FEL23]**

Feature attribution

CONCEPT-BASED INTERPRETABILITY

Assigning importance scores to input features or internal variables for a model prediction. Concept methods address a different level of explanation by grouping behavior into higher-level directions or concepts.

▶ **Sources [GHO19] [FEL23]**

Concept attribution

CONCEPT-BASED INTERPRETABILITY

Estimating how a higher-level candidate concept contributes to a prediction. CRAFT combines concept discovery with concept importance and concept attribution maps.

▶ **Sources [FEL23]**

CAV

CONCEPT-BASED INTERPRETABILITY

A Concept Activation Vector: a direction in activation space associated with a concept.

▶ **Interpretation boundary**

▶ **Sources [KIM18] [FEL23]**

TCAV

CONCEPT-BASED INTERPRETABILITY

Testing with Concept Activation Vectors uses directional sensitivity in activation space to quantify relevance of user-defined concepts to predictions. It provides historical context for concept directions before automatic concept discovery methods such as ACE and CRAFT.

▶ **Sources [KIM18]**

ACE

CONCEPT-BASED INTERPRETABILITY

Automatic Concept-based Explanations automatically extracts visual concepts and evaluates their relevance to predictions. ACE is a key step from user-specified concepts toward automatic concept discovery.

▶ **Sources [GHO19]**

CRAFT

CONCEPT-BASED INTERPRETABILITY

Concept Recursive Activation FacTorization for Explainability: an automatic concept method using nonnegative activation factorization, recursive decomposition, Sobol-based importance, and concept attribution maps. It is the main interpretability case study in Lab 4.

▶ **Sources [FEL23]**

Image crop / patch

CONCEPT-BASED INTERPRETABILITY

A local region extracted from an image and analyzed as an example. CRAFT-style concept discovery uses crop activations so that candidate concepts can be associated with recurring visual parts or patterns.

▶ **Sources [FEL23]**

Global average pooling

CONCEPT-BASED INTERPRETABILITY

Averaging a convolutional activation map over spatial positions to obtain one feature value per channel. CRAFT uses pooled crop activations for its NMF concept-factorization stage.

▶ **Sources [FEL23]**

Concept bank

CONCEPT-BASED INTERPRETABILITY

A collection of candidate concept directions used to represent or analyze activations.

▶ **Interpretation boundary**

▶ **Sources [FEL23]**

Concept usage / coefficient

CONCEPT-BASED INTERPRETABILITY

A coefficient describing how strongly a candidate concept contributes to one observation's representation. In $A \approx UW^\top$, U_{ij} is the usage of candidate j by crop i.

▶ **Sources [FEL23]**

Concept importance

CONCEPT-BASED INTERPRETABILITY

How strongly model behavior depends on a candidate concept, rather than merely whether the concept is present.

▶ **Why it matters**

▶ **Sources [FEL23]**

Sobol sensitivity

CONCEPT-BASED INTERPRETABILITY

A variance-based global sensitivity measure that attributes output variation to input variables or groups. CRAFT applies Sobol indices to concept importance; simplified notebook meters should be labeled as pedagogical proxies unless they reproduce the full estimator.

▶ **Sources [FEL23]**

Concept Attribution Map

CONCEPT-BASED INTERPRETABILITY

A spatial map localizing evidence associated with a candidate concept in an input. It matters when comparing CRAFT's pooled NMF stage with tensor extensions that retain location directly.

▶ **Sources [FEL23]**

Recursive concept decomposition

CONCEPT-BASED INTERPRETABILITY

Re-analyzing a higher-level concept using activations from an earlier layer to obtain finer sub-concepts. It illustrates that the semantic granularity of a concept depends on the layer.

▶ **Sources [FEL23]**

Concept granularity

CONCEPT-BASED INTERPRETABILITY

The level of semantic detail expressed by a concept, from class-level structure to parts, textures, or other sub-concepts. Rank, layer, and representation family can all change the concept vocabulary and its granularity.

▶ **Sources [FEL23]**

IV. Low-rank learning in neural networks

LoRA

LOW-RANK LEARNING IN NEURAL NETWORKS

Low-Rank Adaptation represents a trainable weight update with a low-rank parameterization rather than updating all pretrained weights. It motivates questions about whether learning factors, subspaces, or the fixed-rank object is the right geometric viewpoint.

▶ **Sources [HUE22] [LIS25]**

Adapter

LOW-RANK LEARNING IN NEURAL NETWORKS

A small trainable module or parameter update inserted into or associated with a pretrained model for parameter-efficient adaptation. LoRA is a low-rank adapter strategy.

▶ **Sources [HUE22]**

Low-rank weight update

LOW-RANK LEARNING IN NEURAL NETWORKS

A weight change constrained to have low matrix rank. StelLA decomposes such an update as $USV^\top$ to make input/output subspaces explicit.

▶ **Sources [LIS25]**

Subspace learning

LOW-RANK LEARNING IN NEURAL NETWORKS

Learning the low-dimensional space in which useful updates or representations live, rather than assigning meaning to one particular basis. StelLA explicitly learns orthonormal input and output subspaces using Stiefel factors.

▶ **Sources [LIS25]**

Factorized parameterization

LOW-RANK LEARNING IN NEURAL NETWORKS

Representing a low-rank object through factors, e.g. $W = AB^\top$. It reduces parameter count but introduces non-unique coordinates and can affect conditioning.

▶ **Sources [BIA25]**

Parameter redundancy

LOW-RANK LEARNING IN NEURAL NETWORKS

Different parameter values encode the same represented object or model behavior. RAdaGrad/RAdamW motivate direct fixed-rank optimization partly to avoid redundant factor coordinates.

▶ **Sources [BIA25]**

Fixed-rank weight

LOW-RANK LEARNING IN NEURAL NETWORKS

A matrix parameter constrained to have a specified rank. It can be optimized as an intrinsic matrix object rather than only through a factorization.

▶ **Sources [BIA25]**

RAdaGrad / RAdamW

LOW-RANK LEARNING IN NEURAL NETWORKS

Riemannian optimizers for fixed-rank matrix weights that adapt AdaGrad/AdamW-style metrics to the fixed-rank manifold. They illustrate geometry-aware low-rank learning without relying on a redundant factorization.

▶ **Sources [BIA25]**

Equivariance

LOW-RANK LEARNING IN NEURAL NETWORKS

A property where transforming the input induces a predictable corresponding transformation of the representation or output. Tensor Decomposition Networks study low-rank approximations inside SO(3)-equivariant architectures.

▶ **Sources [LIN25]**

Tensor product

LOW-RANK LEARNING IN NEURAL NETWORKS

A multilinear construction that combines representation spaces or features. Equivariant networks may use structured tensor-product interactions that can be expensive.

▶ **Sources [LIN25]**

Clebsch–Gordan tensor product

LOW-RANK LEARNING IN NEURAL NETWORKS

A structured tensor-product operation used to combine SO(3)-equivariant features while respecting representation-theoretic symmetry. TDNs replace expensive CG tensor-product operators with low-rank tensor decompositions.

▶ **Sources [LIN25]**

Low-rank tensor operator

LOW-RANK LEARNING IN NEURAL NETWORKS

A tensor-valued or multilinear operator approximated using a low-rank tensor decomposition. TDNs use CP structure to reduce the cost of tensor-product operations.

▶ **Sources [LIN25]**

Architectural compression

LOW-RANK LEARNING IN NEURAL NETWORKS

Reducing computation or parameter cost by replacing a large operator or module with a structured lower-complexity approximation. In TDNs the low-rank tensor structure changes the architecture's operator implementation, not merely post-hoc interpretation.

▶ **Sources [LIN25]**

Separable interaction

LOW-RANK LEARNING IN NEURAL NETWORKS

An interaction that factors into products of simpler mode-wise terms. CP decomposition imposes a separability assumption on each rank-one component.

▶ **Sources [KB09] [LIN25]**

V. Evidence and validation

Mechanistic interpretability

EVIDENCE AND VALIDATION

Studying internal model computations with the goal of identifying features, circuits, or mechanisms that explain behavior. SAE feature dictionaries are one current tool in this broader program.

▶ **Sources [CUN23] [BRI23]**

Activation patching

EVIDENCE AND VALIDATION

Replacing or restoring internal activations from another run to test whether a particular activation state contributes causally to behavior. It is an intervention-style diagnostic distinct from merely observing a large feature activation.

Ablation

EVIDENCE AND VALIDATION

Removing, zeroing, or disabling a feature, unit, component, or pathway and measuring the resulting behavioral change. It can provide causal evidence when appropriate controls rule out trivial distribution-shift effects.

Intervention

EVIDENCE AND VALIDATION

Deliberately changing a proposed mechanism or representation and testing whether the predicted effect follows. It is treated as stronger evidence than reconstruction or correlation alone.

Causal relevance

EVIDENCE AND VALIDATION

Evidence that changing a candidate feature or mechanism changes model behavior in the predicted way. A feature can be present or correlated without being causally important.

▶ **Sources [CUN23]**

Faithfulness

EVIDENCE AND VALIDATION

The degree to which an explanation accurately reflects the model computation or behavior it claims to explain. CRAFT evaluates concept-importance faithfulness; visual plausibility alone is not enough.

▶ **Sources [FEL23]**

Semantic coherence

EVIDENCE AND VALIDATION

The extent to which examples strongly associated with a learned feature support a consistent human interpretation. Coherence supports naming a candidate concept but does not alone establish causal importance.

▶ **Sources [GHO19] [FEL23]**

Stability

EVIDENCE AND VALIDATION

Persistence of a candidate structure across reasonable changes in initialization, samples, ranks, perturbations, or solver choices.

▶ **Interpretation boundary**

Feature universality

EVIDENCE AND VALIDATION

The hypothesis or observation that similar learned features recur across models, scales, seeds, or training settings. It can strengthen evidence that a feature is not a one-run artifact, but matching features across representations requires care.

Held-out validation

EVIDENCE AND VALIDATION

Testing a claim on data that were not used to fit or select the representation. A semantic label should generalize beyond the examples used to invent that label.

Behavioral validation

EVIDENCE AND VALIDATION

Testing whether a proposed concept is meaningfully related to model outputs on external examples or tasks. It separates 'this factor looks interpretable' from 'this factor predicts something about model behavior'.

▶ **Sources [KIM18] [GHO19] [FEL23]**

Semantic validation

EVIDENCE AND VALIDATION

Evidence that a mathematical factor corresponds to the domain meaning assigned to it. High-usage examples can suggest a label; stability, held-out behavior, or interventions strengthen the claim.

Reconstruction

EVIDENCE AND VALIDATION

How accurately a learned representation reproduces the data or activation object it was fit to.

▶ **Interpretation boundary**

Alignment-relevant evidence

EVIDENCE AND VALIDATION

Representation- or behavior-level evidence that may support auditing or monitoring questions relevant to system behavior. The tutorial does not present tensor decomposition as an alignment algorithm; it studies when low-rank evidence is sufficiently invariant, stable, and behaviorally validated to be useful for auditing.

Cross-language map

A useful way to read the tutorial is to translate a mathematical object into the interpretability claim someone may want to make about it.

Mathematical language	Interpretability language	Caution
factor / dictionary atom	candidate feature or concept direction	a direction becomes semantic only after validation
coefficient	feature or concept usage	presence is not importance
subspace	representation or update subspace	a subspace can be stable even when a particular basis is arbitrary
rank	vocabulary capacity / granularity	more rank can split or duplicate concepts
conditioning	representation stability	good reconstruction can coexist with fragile coordinates
identifiability	recoverability	recoverability is not domain meaning
perturbation / intervention	behavioral evidence	interpretation depends on the intervention and controls

A mathematically recoverable factor is not automatically a scientifically meaningful concept.

Compare representation families

Method	Learned representation	Main structural assumption	Interpretability caution
SVD / PCA-style	orthogonal directions / subspace	orthogonality, variance structure	basis directions need not be semantic
NMF	nonnegative directions + nonnegative usage	additive nonnegative mixture	nonnegative ≠ unique or meaningful
Dictionary learning	reusable atoms + coefficients	learned basis, often sparse coding	atom ≠ concept
Sparse autoencoder	decoder features + sparse latent code	nonlinear encoder + sparse feature usage	SAE feature ≠ validated concept
CPD	rank-one tensor components	separability across modes	scaling/permutation and stability matter
Tucker	mode subspaces + interaction core	low multilinear rank	core coordinates are basis-dependent
BTD	multilinear-rank blocks	structured within-component interaction	block semantics require validation

Sparse autoencoder mini-reference

A sparse autoencoder learns

$$z = f_{enc}(x), \qquad \hat{x} = f_{dec}(z),$$

with a reconstruction objective plus a sparsity-inducing term,

$$\mathcal{L} = \|x - \hat{x}\|^2 + \lambda\,\Omega(z).$$

The **encoder** determines which latent features activate. The **decoder** maps those latent coefficients back into activation space. Decoder directions are often interpreted as learned features or dictionary atoms.

The important boundary for this tutorial is:

$$\textbf{SAE feature} \neq \textbf{validated semantic concept.}$$

To make the semantic step, inspect coherence, stability, held-out behavior, and controlled interventions.

Reference map

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